The Open Source Audit

A Capstone Project for OSS NGMC Course

Course: Open Source Software | Unit Coverage: 1 – 5
Max Marks: 100

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OPEN SOURCE SOFTWARE ANALYSIS REPORT — PYTHON

Introduction

- Open source software has really changed how we make and use software over the world. It is different from software that is owned by a company. Open source software lets people look at the code make changes and share it with others. This is good because it helps people work together and come up with ideas.

- This report is about **Python**, a popular open source programming language. **Python** is used for things like making websites, artificial intelligence looking at data, machine learning and automating tasks. **Python** is easy to understand and use so people who are just starting out and experienced developers all like it.

- The report talks about how **Python** was made what it is about how it is licensed and why it is important, in the Linux world. It also compares open source software to software that is owned by a company. We even. Ran five shell scripts to show that we understand how to use Linux commands and write shell scripts.

Part A. Origin and Philosophy

A1. Problem Python Was Designed to Solve

- Python was made by Guido van Rossum in 1991. Then languages like C and C++ were very strong but hard to understand and use. Even simple programs needed a lot of work and lines of code.

- Python was designed to fix these issues by doing a things. For example it:

  * provides a simple and easy to understand way of writing code
  * makes the code less complicated
  * helps people who're new to programming
  * helps developers get more work done

- The people who made Python believe that code should be simple and easy to read. They think that code should be easy to write. This idea is very important to Python. You can see it in the way Python is designed. Of using brackets Python uses indentation to make the code look neat and organized. This makes Python code easy to read and write. Python is about making code simple and easy to understand. The goal of Python is to make programming easy, for everyone, including people who are just starting out with Python.

A2. License Analysis

- Python is available under the Python Software Foundation License. This license is open to everyone. It is not restrictive.

- The Main Points Of Python -

People who use Python have a lot of freedom. They can do the following things with Python:

1. They can use Python for anything they want
2. They can look at how Python works and try to understand it
3. They can make changes to Python
4. They can give Python to people.

Commercial Use

Python is really useful for things and you do not have to pay any money to use it. A lot of companies use Python when they make their products and services.
GPL vs MIT Comparison

- GPL License is like this: if you change something you have to keep it open for other people to use
- MIT License is more relaxed and does not have a lot of rules
- Python has a license that's similar to the MIT License so you can use it for open source things or for closed source things.
- Free, vs Freedom
- Python is free which means it does not cost any money. It also gives you the freedom to change it and share it with other people. You can do what you want with Python because it is free.

Part B -Linux Footprint

Installation of Python

- To check if Python is installed:
- python3 –version
- If not installed:
- sudo apt update
- apt install python3

Key Directories

- Python is usually installed in the following directories:
- /usr/bin/python3 → Executable file
- /usr/lib/python3.x/ → Libraries
- To locate Python:
- which python3

User and Permissions

- To identify the currently logged-in user in Linux, the following command can be used:
- Whoami
- Linux manages access to files, directories, and programs through a permission system. When Python programs are executed, they also follow these Linux permission rules to determine whether a user is allowed to run or modify them.

Updating Python

- The following command is used to update the system packages, including Python:
- sudo apt update && sudo apt upgrade
- Running this command refreshes the package list and installs the latest available updates for Python and other software packages in the system.

Part C. FOSS Ecosystem

- The Python community is really big. People work on it all the time to make it better. People from over the world help with this.
- Community Contribution.
- People who make software help with Python on websites like GitHub. They do a lot of things to help including:
  • Telling others about problems with the software
  • Suggesting ideas to make Python better
  • Adding new code to make Python work better

Popular Python Libraries

- There are some Python libraries that a lot of people use, including:
  • NumPy → This is used for things that need numbers and science
  • Pandas → This helps people work with data and understand it
  • TensorFlow→ This is used for machine learning and artificial intelligence which's, like making computers smart

Impact of the Community

- The people who work on Python all around the world are very important. They help make Python better by fixing problems making it safer and adding things that it can do. The Python community is always working together to make Python the best it can be. The Python community does a job of helping Python by contributing to it all the time

Part D — Comparison with Proprietary Software

Feature	Python (Open Source)	Proprietary Software
Cost	Free to use	Usually requires payment or license
Source Code	Publicly available	Not accessible to users
Flexibility	Highly flexible	Limited flexibility
Customization	Users can modify and adapt the code	Customization is generally restricted
Support	Supported by the global developer community	Supported by the company that owns the software

Analysis

- Open-source software such as Python provides greater flexibility because users can access and modify the source code according to their needs. It is also free to use and benefits from community support. In contrast, proprietary software is controlled by a specific company, often requires payment, and does not allow users to view or change the source code.

Shell Script Implementation

The following shell scripts were created and executed on a Linux system.

Script 1 — System Identity Report

Script Code

```
#!/bin/bash
# Script 1: System Identity Overview
# Author: Chandraprakash (24BAI10787)

name="Chandraprakash"
reg_no="24BAI10787"
fav_tool="Python"

# Collect system details
kernel_version=$(uname -r)
architecture=$(uname -m)
current_user=$(whoami)
home_path=$HOME
shell_used=$SHELL
uptime_info=$(uptime -p)
today=$(date)

# OS detection
if [[ -f /etc/os-release ]]; then
    os_name=$(grep "PRETTY_NAME" /etc/os-release | cut -d
"" -f2)
else
    os_name="Not Available"
fi

echo "----------------------------------------"
echo " Open Source Report - $name ($reg_no)"
echo "----------------------------------------"
echo "OS        : $os_name"
echo "Kernel    : $kernel_version"
echo "Arch      : $architecture"
echo ""
echo "User      : $current_user"
echo "Home Dir  : $home_path"
echo "Shell     : $shell_used"
echo ""
echo "Uptime    : $uptime_info"
```

```
echo "Date      : $today"
echo ""
echo "License   : GPL"
echo "Tool Used : $fav_tool"
echo "----------------------------------------"
```

Code Output –

```
chandraprakash@LAPTOP-6G17RIID: ~
To run a command as administrator (user "root"), use "sudo <command>".
See "man sudo_root" for details.

chandraprakash@LAPTOP-6G17RIID:~$ nano script1.sh
chandraprakash@LAPTOP-6G17RIID:~$ chmod +x script1.sh
chandraprakash@LAPTOP-6G17RIID:~$ ./script1.sh
----------------------------------------
 Open Source Report - Chandraprakash (24BAI10787)
----------------------------------------
OS         : Ubuntu 24.04.4 LTS
Kernel     : 6.6.87.2-microsoft-standard-WSL2
Arch       : x86_64

User       : chandraprakash
Home Dir   : /home/chandraprakash
Shell      : /bin/bash

Uptime     : up 1 minute
Date       : Sun Mar 29 11:37:07 UTC 2026

License    : GPL
Tool Used  : Python
----------------------------------------
chandraprakash@LAPTOP-6G17RIID:~$
```

Explanation

This script displays system-related information using Linux commands and command substitution

Script 2 — Python Package Inspector

Script Code

```
#!/bin/bash
# Script 2: Detailed Package Inspector
# Author: Chandraprakash (24BAI10787)

pkg="python3"

echo "========== PACKAGE REPORT ==========="
```

```
echo "Package : $pkg"
echo "-----------------------------------"

if dpkg -s "$pkg" &>/dev/null; then
   echo "Status      : Installed"

   version=$(dpkg -s "$pkg" | grep Version | awk '{print $2}')
   arch=$(dpkg -s "$pkg" | grep Architecture | awk '{print $2}')

   echo "Version     : $version"
   echo "Architecture: $arch"

   if command -v python3 >/dev/null; then
      echo "Path        : $(which python3)"
      echo "Interpreter : $(python3 --version)"
   fi

   echo ""
   echo "Quick Info:"
   echo "Python is an open-source language focused on simplicity."

else
   echo "Status  : Not Installed"
   echo "Command : sudo apt install $pkg"
fi

echo "-----------------------------------"
echo "Checked on: $(date)"
```

Output Screenshot

```
chandraprakash@LAPTOP-6G17RIID: ~
chandraprakash@LAPTOP-6G17RIID:~$ nano script2.sh
chandraprakash@LAPTOP-6G17RIID:~$ chmod +x script2.sh
chandraprakash@LAPTOP-6G17RIID:~$ ./script2.sh
========== PACKAGE REPORT ==========
Package : python3
-----------------------------------
Status      : Installed
Version     : 3.12.3-0ubuntu2.1
Architecture: amd64
Path        : /usr/bin/python3
Interpreter : Python 3.12.3

Quick Info:
Python is an open-source language focused on simplicity.
-----------------------------------
Checked on: Sun Mar 29 11:41:54 UTC 2026
chandraprakash@LAPTOP-6G17RIID:~$
```

Explanation

This script checks Python installation status and displays package details.

Script 3 — Disk and Permission

Auditor Script Code

```
#!/bin/bash
# Script 3: Directory & Disk Analyzer
# Author: Chandraprakash (24BAI10787)

folders=("/etc" "/var/log" "/home" "/tmp")

echo "========== Directory Report ==========="
echo ""

for f in "${folders[@]}"; do
    if [[ -d "$f" ]]; then
        details=$(ls -ld "$f")
        perm=$(echo $details | awk '{print $1}')
        owner=$(echo $details | awk '{print $3}')
        size=$(du -sh "$f" 2>/dev/null | cut -f1)

        echo "Path        : $f"
        echo "Permissions : $perm"
        echo "Owner       : $owner"
        echo "Size        : $size"
        echo "-----------------------------------"
    else
        echo "$f -> Not Found"
    fi
done

echo ""
echo "Disk Snapshot:"
df -h | head -n 5

echo ""
echo "Completed on: $(date)"
```

Output Screenshot

```
chandraprakash@LAPTOP-6G17RIID: ~
chandraprakash@LAPTOP-6G17RIID:~$ nano script3.sh
chandraprakash@LAPTOP-6G17RIID:~$ chmod +x script3.sh
chandraprakash@LAPTOP-6G17RIID:~$ ./script3.sh
========== Directory Report ==========

Path        : /etc
Permissions : drwxr-xr-x
Owner       : root
Size        : 4.3M
-----------------------------------
Path        : /var/log
Permissions : drwxrwxr-x
Owner       : root
Size        : 33M
-----------------------------------
Path        : /home
Permissions : drwxr-xr-x
Owner       : root
Size        : 56K
-----------------------------------
Path        : /tmp
Permissions : drwxrwxrwt
Owner       : root
Size        : 24K
-----------------------------------

Disk Snapshot:
Filesystem      Size  Used Avail Use% Mounted on
none            3.9G     0  3.9G   0% /usr/lib/modules/6.6.87.2-microsoft-standard-WSL2
none            3.9G  4.0K  3.9G   1% /mnt/wsl
drivers         453G  131G  322G  29% /usr/lib/wsl/drivers
/dev/sdd       1007G  1.4G  955G   1% /

Completed on: Sun Mar 29 11:43:22 UTC 2026
chandraprakash@LAPTOP-6G17RIID:~$
```

Explanation

This script checks directory permissions and disk usage using loops.

Script 4 — Log File Analyzer

Script Code

```
#!/bin/bash
# Script 4: Log Scanner
# Author: Chandraprakash (24BAI10787)

if [[ $# -lt 1 ]]; then
    echo "Usage: $0 <file> [search_word]"
    exit 1
fi

file=$1
search=${2:-"error"}

if [[ ! -f "$file" ]]; then
    echo "File does not exist!"
    exit 1
fi

total_lines=$(wc -l < "$file")
count=0

while read line; do
    echo "$line" | grep -iq "$search" && ((count++))
done < "$file"

echo "========== Log Summary ==========="
echo "File       : $file"
echo "Keyword    : $search"
echo "Total Line : $total_lines"
echo "Matches    : $count"

if [[ $total_lines -gt 0 ]]; then
    percent=$(echo "scale=2; ($count/$total_lines)*100" | bc)
    echo "Match %    : $percent%"
fi

echo ""
echo "Recent Matches:"
grep -in "$search" "$file" | tail -5

echo ""
echo "Done at: $(date)"
```

Output Screenshot

```
chandraprakash@LAPTOP-6G17RIID: ~
chandraprakash@LAPTOP-6G17RIID:~$ nano script4.sh
chandraprakash@LAPTOP-6G17RIID:~$ chmod +x script4.sh
chandraprakash@LAPTOP-6G17RIID:~$ ./script4.sh
Usage: ./script4.sh <file> [search_word]
chandraprakash@LAPTOP-6G17RIID:~$ nano test.txt
chandraprakash@LAPTOP-6G17RIID:~$ ./script4.sh test.txt error
========== Log Summary ==========
File       : test.txt
Keyword    : error
Total Line : 6
Matches    : 3
Match %    : 50.00%

Recent Matches:
1:Error: something failed
4:error occurred again
5:./script4.sh test.txt error

Done at: Sun Mar 29 11:45:58 UTC 2026
chandraprakash@LAPTOP-6G17RIID:~$
```

Explanation

This script counts the number of error messages in a log file.

Script 5 — Open Source Manifesto Generator

Script Code

```
#!/bin/bash
# Script 5: Open Source Pledge Generator
# Author: Chandraprakash (24BAI10787)

clear
echo "===== Open Source Pledge Creator ====="
echo ""

read -p "Enter your name: " uname
read -p "Tool you use daily: " tool
read -p "Meaning of freedom: " freedom
read -p "What will you create? " create

uname=${uname:-"Chandraprakash"}
today=$(date +"%d-%m-%Y")

file_name="pledge_${uname// /_}.txt"

cat <<EOF > $file_name
----------------------------------------
 OPEN SOURCE PLEDGE - $uname
----------------------------------------
Date: $today

I, $uname, support open-source principles.

Using $tool daily shows how collaboration works.

Freedom means: $freedom

I promise to build: $create
and share it with everyone.

Signed,
$uname
($today)
----------------------------------------
EOF

echo ""
echo "File saved as: $file_name"
echo ""
cat $file_name
```

Output Screenshot

```
chandraprakash@LAPTOP-6G17RIID: ~
===== Open Source Pledge Creator =====

Enter your name: Chandraprakash
Tool you use daily: java
Meaning of freedom: feel free
What will you create? a file

File saved as: pledge_Chandraprakash.txt

----------------------------------------
 OPEN SOURCE PLEDGE - Chandraprakash
----------------------------------------
Date: 29-03-2026

I, Chandraprakash, support open-source principles.

Using java daily shows how collaboration works.

Freedom means: feel free

I promise to build: a file
and share it with everyone.

Signed,
Chandraprakash
(29-03-2026)
----------------------------------------
chandraprakash@LAPTOP-6G17RIID:~$
```

Explanation

This script takes user input and saves it into a file.

Conclusion

- Python is a highly versatile and powerful open-source programming language that is widely used in many fields. Its simple syntax and strong global community make it an excellent choice for both beginners and experienced developers.
- Through this project, both theoretical concepts and practical aspects of open-source software and the Linux environment were explored. The shell scripting exercises helped

develop an understanding of automation, file management, and basic system operations.

- In conclusion, Python and open-source technologies have become an essential part of modern computing. With continuous community contributions and technological advancements, their importance and usage are expected to grow even further in the future.